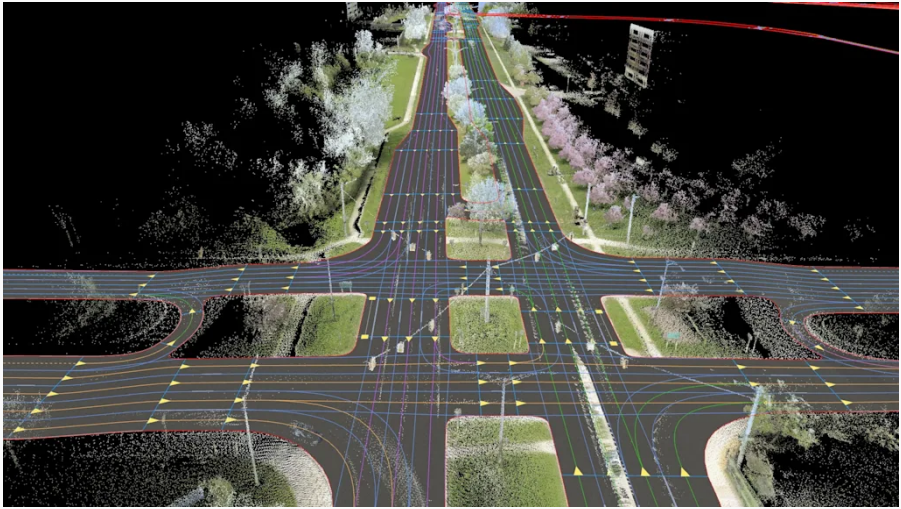
HD Maps

Michael Schweitzer

June 16, 2026

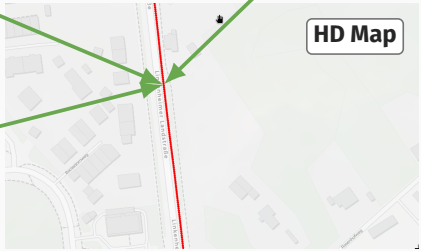
HD Maps (here)



HD Map Localization



sensor data
EGO
IMU
Stereo
...



IBM Rational DOORS — Formal module: 'HD Map Requirements' — /HD Map/Localization

File Edit View Insert Link Analysis Table Tools Discussions Change User Help

View: Standard view

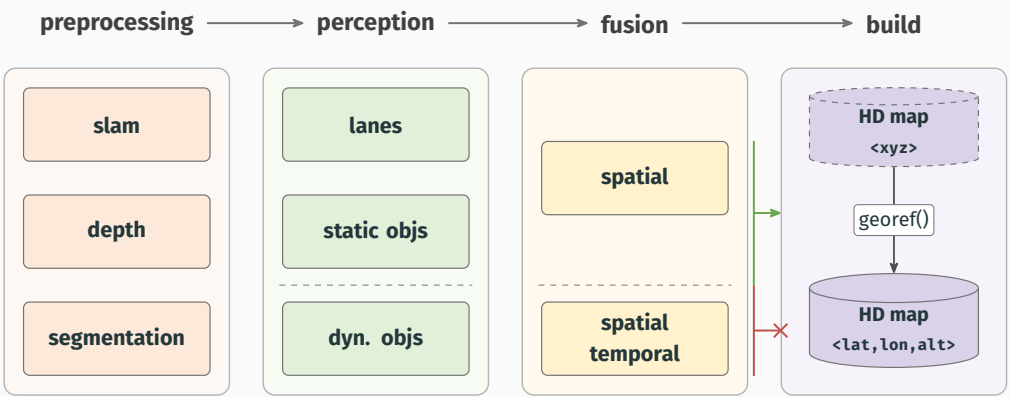
#	ID	Object Text
2		**2 Localization requirements**
2.1	**HD-001**	The map-relative accuracy error shall not exceed 0.1 m.
2.2	HD-002	The localization estimate shall be available at 10 Hz.
2.3	HD-003	The localization latency shall not exceed 50 ms.

Username: mschweitzer — Exclusive edit mode — 1 object selected

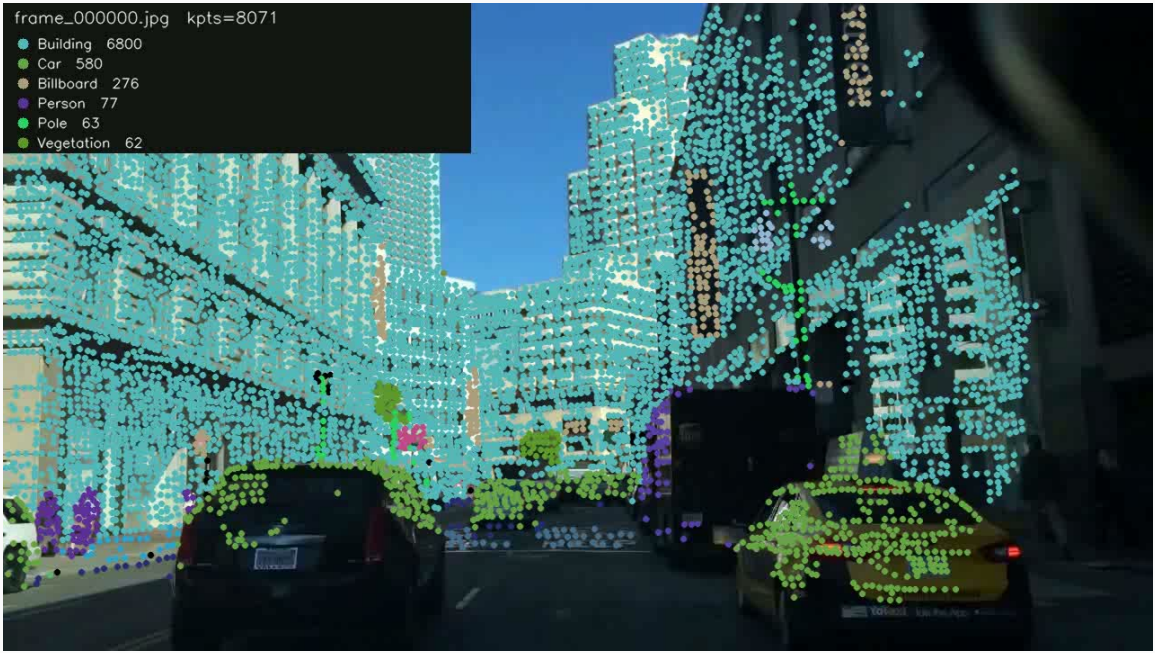
Outline

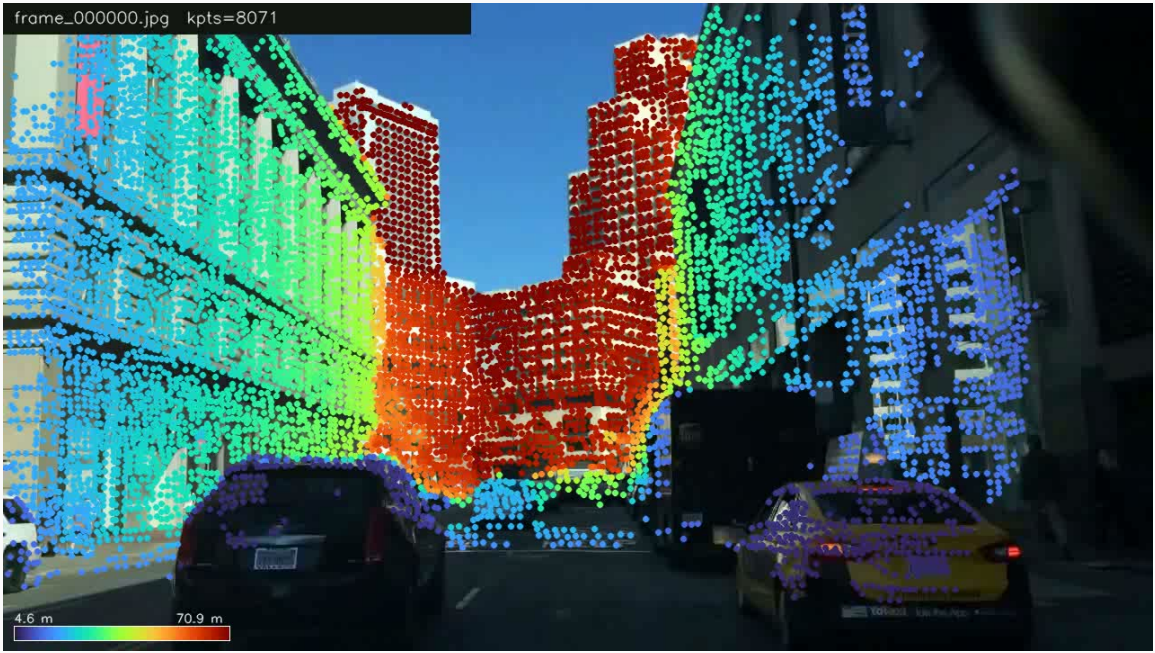
- Introduction ✓ *(already done!)*
- HD Map Building *(prototype)*
- HD Map Localization *(prototype)*
- Conclusion
- Questions

HD Map Building Pipeline (*Prototype*)




perception cache v1 step=3
frame 000000 / 000099 npz: yes
semantic
instance
mono_depth
depth_conf

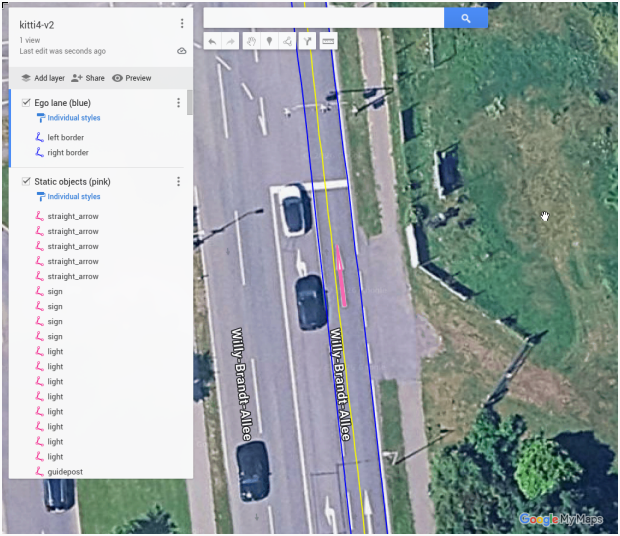

frame_000000.jpg kpts=8071
Building 6800
Car 580
Billboard 276
Person 77
Pole 63
Vegetation 62


frame_000000.jpg kpts=8071
4.6 m
70.9 m

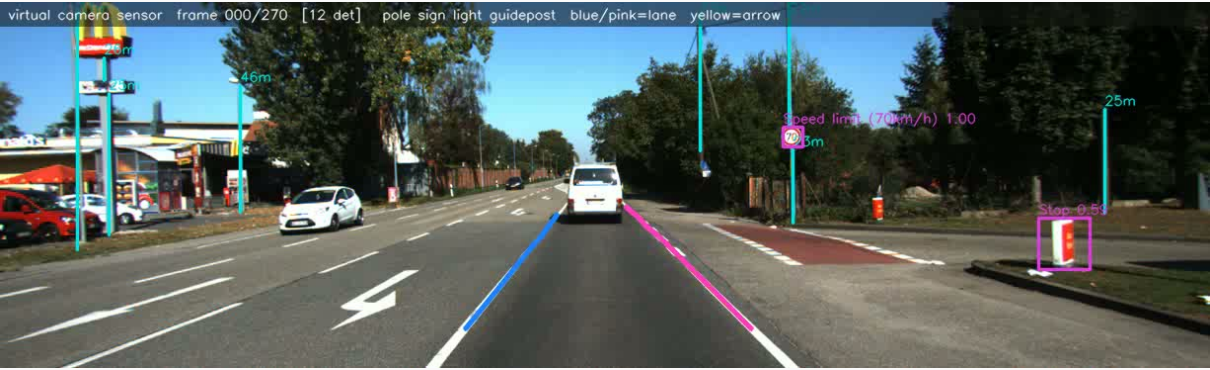

frame 000/270 ego-lane=green border=black | yellow=arrow magenta=sign green=light white=guidepost | red=moving orange=pot_dyn (in-box) (poles excluded)
car_pot_dyn
car_DYN
car_DYN
A0
70
70



HD Map Building - `georef()`

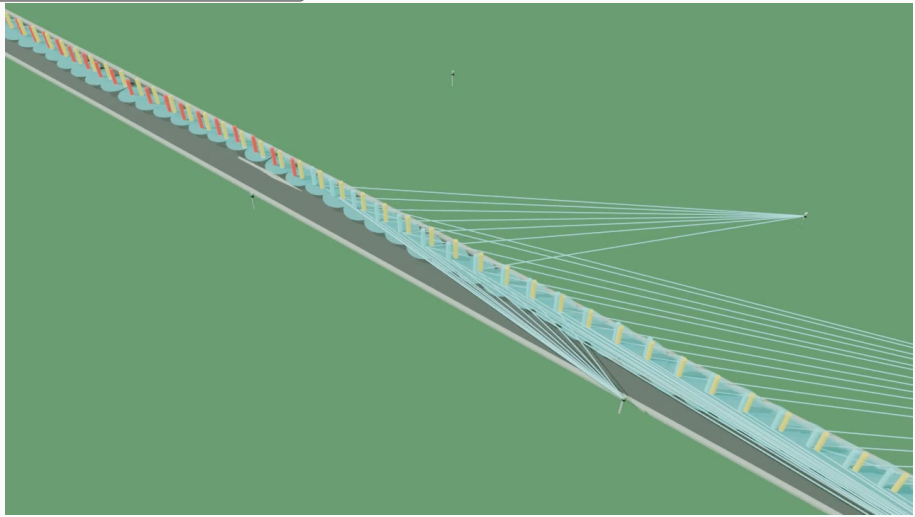


HD Map Localization (Prototype): Virtual Camera



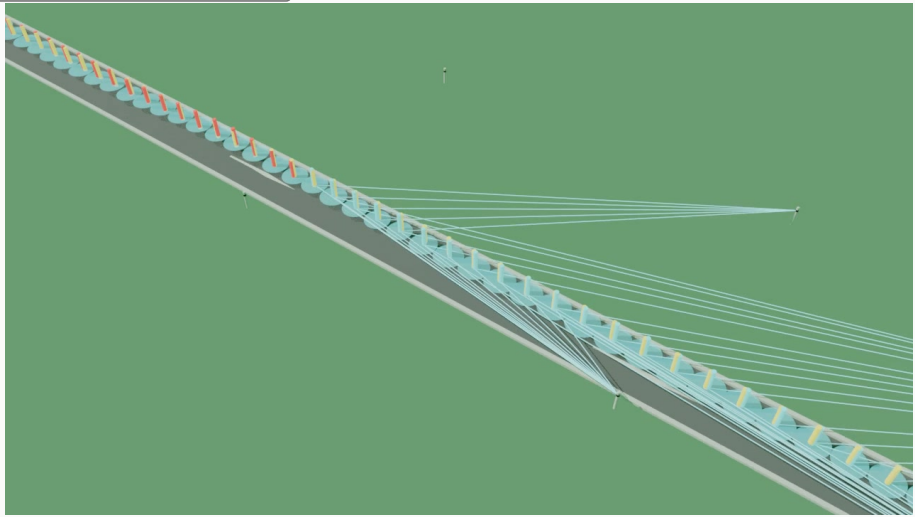
HD Map Localization (Prototype): Factor Graph

vcam pixel noise = 10 px



HD Map Localization (Prototype): Factor Graph

vcam pixel noise = 0 px



HD Map Localization – Accuracy

	0 px (before)	10 px (now)
ATE East/North	0.255 m	0.589 m

$$\mathrm{ATE}_{EN} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left[(E_i^{\mathrm{est}} - E_i^{\mathrm{gt}})^2 + (N_i^{\mathrm{est}} - N_i^{\mathrm{gt}})^2\right]}$$

ATE = Absolute Trajectory Error

Conclusion

Questions?